THE2 Report

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I. Reading and saving the images

The reading part is the same as last time, but for this assignment, due to the variety of methods involved we ended up foregoing a universal write function in favor of printing the images within the functions for the Transforms part of the assignment. (Also because we were rather confused about what kind of image we were trying to get.)

II. Transformations

A. Fourier

This transformation was provided as an example in the lecture notes, albeit for a grayscale image. Applying the transform to R,G and B bands seperately was relatively easy, but combining them together was where it got confusing. In the end, for the fourier transform, I decided to display them seperately on grayscale, because they matched the fourier transforms displayed by the lecture notes, and because there was more to see by the human eye. Notably, before transitioning to the Stretched Spectrum, the figures were pretty much pitch black, which set a harrowing precedent of not knowing what to look for.

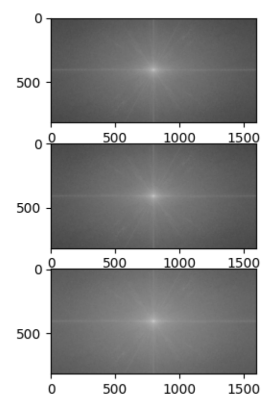

Fig. 1. Image 1, Fourier Transforms

B. Cosine

For this transform, we ran into the opposite issue where displaying the bands seperately resulted in nothing to observe. Stacking them together and then normalizing the colors did produce a noisy image that vaguely resembled the figures in Lecture Notes in that the upper left corner got busy, but I found that unsatisfying. So we left the seemingly blank outputs as is since calling the inverse function on them did indeed produce the original image, so there must have been something in there that human eye can't see.

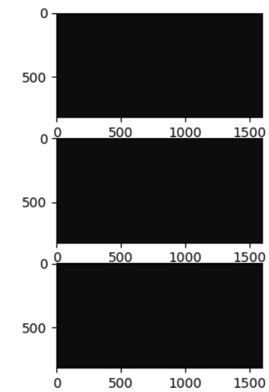

Fig. 2. Image 2, Cosine Transforms?

C. Hadamard

This transformation required a square-shaped image that is also the length of power 2, which required some calculations and cropping. And after all that, the output was yet again a seemingly pitch-black rectangle. Which was really confusing, and hard to confirm the validity of given the precedent set by the prior two transforms. It also takes exceedingly long to print, which is not encouraging.

III. Filters

This is where we switched to a universal write-image function that normalized the RGB values as requested, which produced some truly horrifying vistas that stunned us as we tried to confirm if we did something wrong. Otherwise, there wasn't much difference in implementing high and low passes. The ideal, butter, and gaussian methods were a bit difficult to make sense of (and the example in the Lecture Notes was rather dreadful), but other several online resources ... helped us get to those horrifying vistas. For example, I'm genuinely unsure if this butterworth high pass filter is supposed to be

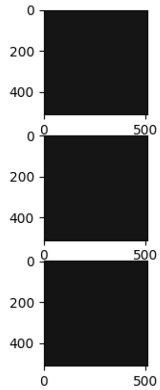

Fig. 3. Image 3, My terrible son

like this, if I terribly failed the color normalization, or if my eyes are malfunctioning.

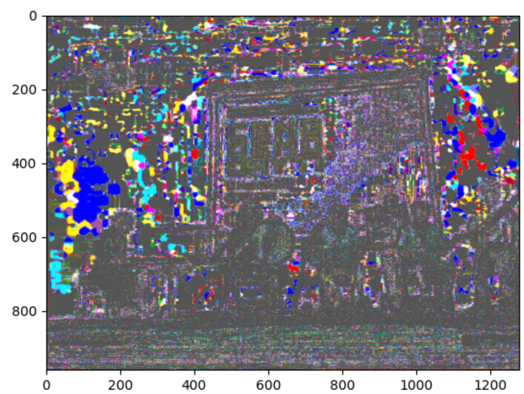

Fig. 4. Image 4, ???

IV. Band Filters

Thankfully straghtforward Low + High pass filters. As for clearing up the image of noise, that would have been a theoretically simple affair of looking at the fourier output and determining the required cutoff frequencies to get rid of the abnormalities. But in reality, the fourier transforms we of the images didn't have any conveniently circular targets, so we had to manually guess the frequency inputs and hope for the best. But in the end the image ended up as more of a mess instead.

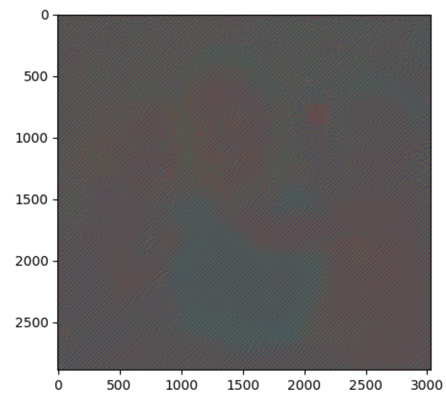

Fig. 5. Image 5, I can't help this person